Computability & Logic, Exercises to Lectures 1+2

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Theoretical Exercises

From John C. Martin book

Advice

The material might be slightly too much for the three hours session, but with a proper preparation you should get quite far. Make sure you do some exercises in each category. The minimal sequence would be 1.31, 1.33, 3.1, 3.2, 3.7(a-c,g), 2.1, 2.2, 2.5 + the extra exercise "Induction over Regular Expressions", which is a good preparation for the obligatory assignment (A1).

The other exercises are also important (esp. 3.9, 2.10), so you can get more out of your exercise class with a proper preparation. In case you have time left, note that the book contains plenty more interesting challenge problems!

Languages:

- 1.31. Mind the proof structure! How do you prove if-and-only-if? How to prove equality of sets $A = B$?
- 1.33 (a-c). Make sure the structure of your proof reflects that you prove a statement of the form $A \subseteq B$.

Regular Expressions:

- 3.1 (a-b)
- 3.2
- 3.7 (a-c,g)
- 3.7 (m) (to show that it is not always easy, this is hard to do with your current tools!)
- 3.9. Why does the same argument not work for infinite sets?
- Give a regular expression $E \neq \emptyset$, for which $\mathcal{L}(E) = \emptyset$.

Induction over Regular Expressions:

- In this exercise, we practice proof by induction using an existential quantifier. Define that "the set A is *non-empty*" by $\exists x\,(x \in A)$. We will consider restricted regular expressions **without** any occurrence of $\emptyset$, i.e., there are only 5 cases of expressions: Λ, a, $E_1\,E_2$, $E_1 + E_2$, and E_1^*.

  Prove with induction over these restricted regular expressions that $\mathcal{L}(E)$ is non-empty. Your proof should clearly distinguish expressions and languages. Also, the steps involving the $\exists$-quantifier should be explicit. Note that this induction proof should have two base cases, and three induction steps.

(Deterministic) Finite Automata:

- 2.1 (a-c,f,g)
- 2.2 (d,e)
- 2.5
- 2.10 (a,b)